Part 1: Weeks 1–4 (Questions 1–20)

Question	Answer
1. Which of the following is NOT a characteristic of a good algorithm? A) Ambiguity B) Finiteness C) Input D) Effectiveness	
2. In a flowchart, which symbol is used to represent a decision or branching point? A) Rectangle B) Oval C) Diamond D) Parallelogram	
3. Which component is responsible for converting Java bytecode into machine-specific code? A) JDK B) JRE C) JVM D) JIT Compiler	
4. What is the correct signature of the main method in Java? A) public void main(String[] args) B) static public void main(String[] args) C) public static void main(String[] args) D) Both B and C	
5. Which data type is used to store a single character in Java and what is its size? A) char, 1 byte B) char, 2 bytes C) String, 2 bytes D) character, 1 byte	
6. Which of the following is a valid identifier in Java? A) 1stVariable B) @myVar C) _value D) class	
7. What is the output of `System.out.println(10 + 20 + "Java");`? A) 1020Java B) 30Java C) Java1020 D) Error	
8. Which operator is used to compare two values for equality in Java? A) = B) == C) <> D) equals()	
9. Which of the following data types is NOT supported in a switch statement in Java (prior to Java 7)? A) int B) char	

Question	Answer
C) long D) byte	
10. What happens if you omit the break statement in a switch case? A) Compilation Error B) The program stops C) It falls through to the next case D) It exits the switch block	
11. Which statement is true about the do-while loop? A) It executes the body at least once B) It checks the condition before execution C) It is an entry-controlled loop D) It cannot contain a break statement	

12. What is the output of the following code?
`int x = 5;`
`System.out.println(x++ + ++x);`
A) 10
B) 11
C) 12
D) 13	

13. Which of the following logic creates an infinite loop? A) for(;;) B) while(true) C) do { } while(1==1); D) All of the above	
14. What is the result of `10 % 3` in Java? A) 3 B) 1 C) 3.33 D) 0	
15. Which keyword is used to skip the current iteration of a loop and proceed to the next? A) break B) return C) continue D) exit	
16. What is the correct syntax for a ternary operator? A) condition ? true_part : false_part B) condition : true_part ? false_part C) condition ? true_part ; false_part D) condition ?? true_part : false_part	
17. Which variable type typically needs to be initialized before it can be used in a calculation? A) Instance variable B) Local variable C) Static variable D) Class variable	

Question	Answer
18. In the expression `boolean b = (5 > 3) && (8 > 5);`, what is the value of b? A) true B) false C) 1 D) 0	
19. Which of the following is NOT a Java primitive data type? A) int B) float C) boolean D) String	
20. What is the default value of an int variable declared as an instance variable? A) null B) 0 C) 1 D) Undefined	

Part 2: Weeks 5-6 (Questions 21-40)

Question

21. How many times will the print statement execute in this code?
`for (int i = 0; i < 3; i++)`
`for (int j = 0; j < 3; j++)`
`System.out.println(i * j);`
A) 3
B) 6
C) 9
D) 12

22. Which statement is used to exit from an outer loop from within an inner loop in Java? A) break B) exit C) goto D) Labeled break (e.g., `break outer;`)

23. What is the output of the following nested loop structure?
`for(int i=1; i<=2; i++) {`
`for(int j=1; j<=2; j++) {`
`if(i==j) continue;`
`System.out.print(i + "" + j + " ");`
`}`
`}`
A) 11 12 21 22
B) 12 21
C) 11 22
D) 21 12

24. In a nested loop (O(n^2)), if the outer loop runs N times and the inner loop runs N times, what is the time complexity?

Question
A) Linear B) Quadratic C) Logarithmic D) Constant
25. Which of the following is correct regarding nested loops? A) You can only nest `for` loops inside `for` loops. B) You cannot nest a `while` loop inside a `for` loop. C) Any type of loop can be nested inside any other type of loop. D) Nesting depth is limited to 3 levels.
26. Which of the following is the correct syntax to declare an integer array of size 5? A) `int arr[5];` B) `int[] arr = new int[5];` C) `int arr = new int[5];` D) `int[] arr = new int(5);`
27. What is the index of the first element in a Java array? A) 1 B) -1 C) 0 D) It depends on user definition
28. Which property is used to find the size of an array in Java? A) `size()` B) `length()` C) `length` D) `getSize()`
29. What is the default value of elements in an array defined as `int[] arr = new int[5];`? A) null B) 0 C) Garbage value D) -1
30. What exception is thrown if you try to access `arr[5]` in an array of size 5? A) ArrayElementNotFoundException B) NullPointerException C) ArrayIndexOutOfBoundsException D) MemoryOverflowException
31. Which syntax properly initializes a multidimensional array? A) `int[][] myNumbers = { {1, 2}, {3, 4} };` B) `int myNumbers[][] = { (1, 2), (3, 4) };` C) `int[][] myNumbers = [ [1, 2], [3, 4] ];` D) `int[] myNumbers = { {1, 2}, {3, 4} };`
32. In Java, is it possible to have a 2D array where rows have different lengths (Ragged Array)? A) No, all rows must be the same length. B) Yes, this is called a Jagged or Ragged array. C) Yes, but only with `double` data types. D) No, that causes a compilation error.

Question
33. Which loop is best suited for iterating through all elements of an array without needing an index? A) `while` B) `do-while` C) Standard `for` loop D) Enhanced `for-each` loop
34. What happens when you do `int[] a = {1,2}; int[] b = a;`? A) A new copy of the array is created. B) `b` refers to the same array object as `a` (Shallow Copy). C) `b` is a deep copy of `a`. D) Compiler error.
35. Which utility class provides methods like `sort()` and `binarySearch()` for arrays? A) `java.util.ArrayTypes` B) `java.util.Collections` C) `java.util.Arrays` D) `java.lang.Array`
36. What is the output of `System.out.println(args.length);` if run without arguments? A) null B) 0 C) 1 D) Error
37. Can you change the size of an array after it has been created? A) Yes, using `resize()`. B) Yes, by assigning a new value to `.length`. C) No, arrays are fixed-size. D) Yes, if it is an `ArrayList`.
38. How do you correctly pass an array to a method? A) `methodName(int[] array)` B) `methodName(array[])` C) `methodName(array)` D) `methodName(array[0])`
39. Which statement correctly creates an anonymous array? A) `new int[] {1, 2, 3}` B) `new int[3] {1, 2, 3}` C) `{1, 2, 3}` (outside declaration) D) `int[] {1, 2, 3}`
40. What is the result of `int[] a = {10, 20}; System.out.println(a);`? A) 10 20 B) {10, 20} C) The memory address (Hashcode) of the array. D) 10

Part 3: Weeks 7-8 (Questions 41-60)

Question

41. What is the output of the following code regarding pass-by-value?
`public void modify(int x) { x = 10; }`
`public static void main(String[] args) {`
`int x = 5;`
`new Test().modify(x);`
`System.out.println(x);`
`}`
A) 10
B) 5
C) 0
D) Compile Error

42. What happens when you compile and run this code involving integer caching?
`Integer a = 127, b = 127;`
`Integer c = 128, d = 128;`
`System.out.println((a == b) + " " + (c == d));`
A) true true
B) false false
C) true false
D) false true

43. Which method will be invoked in this overloading scenario? `method(5)` is called.
`void method(long l) { ... }`
`void method(Integer i) { ... }`
A) `method(long)`
B) `method(Integer)`
C) Ambiguity Error
D) Compile Error

44. What is the result of the following recursive function for `f(5)`? `int f(int n) { return (n==0) ? 0 : n + f(n-1); }` A) 10 B) 15 C) 20 D) StackOverflowError

45. What is the output of this loop with a tricky boolean expression?
`boolean b = false;`
`if (b = true) System.out.print("A");`
`else System.out.print("B");`
A) A
B) B
C) Compile Error
D) No Output

46. Dealing with octal literals: What is printed?
`int i = 010;`
`int j = 07;`
`System.out.println(i + j);`
A) 17
B) 15

Question
C) 107 D) 87
47. Unreachable code detection: Which statement generates a compiler error? A) `if (false) { System.out.println("A"); }` B) `while (false) { System.out.println("A"); }` C) `for(int i=0; i<0; i++) { }` D) None of the above

48. Floating point arithmetic precision:
`double a = 0.1;`
`double b = 0.2;`
`System.out.println(a + b == 0.3);`
A) true
B) false
C) ArithmeticException
D) 0.30000000000000004

49. What is the output regarding static variable scope? `class Test { static int x = 10; public static void main(String[] args) { Test t1 = new Test(); t1.x++; Test t2 = new Test(); System.out.println(t2.x); } }` A) 10 B) 11 C) 0 D) Compile Error
50. Varargs priority: Which call is valid for `void foo(int... x)` and `void foo(int x, int y)`? A) `foo(1)` calls the second method. B) `foo(1, 2)` calls the second method (fixed args). C) `foo(1, 2)` calls the varargs method. D) Ambiguity Error.

51. Array index evaluation order: What is the state of the array after execution?
`int[] arr = {1, 2, 3};`
`int i = 1;`
`arr[i++] = arr[--i] + 5;`
A) {1, 7, 3}
B) {1, 2, 7}
C) {1, 6, 3}
D) {1, 2, 6}

52. Division by zero with doubles: `System.out.println(1.0 / 0);` A) ArithmeticException B) Infinity C) NaN D) 0.0

53. What happens here regarding String immutability?
`String s = "Java";`
`s.concat(" Programming");`
`System.out.println(s);`
A) Java Programming

Question
B) C) Programming D) Null
54. Modulus with negative numbers: What is `-10 % 3`? A) 1 B) -1 C) 2 D) -2
55. Method signature definition: Which of the following is NOT part of a method's signature? A) Method Name B) Parameter Types C) Return Type D) Order of Parameters
56. What is the output of character addition? `System.out.println('A' + 'B');` A) AB B) 131 C) Compile Error D) A B

57. What is the outcome of executing this block?
`{ int x = 5; }`
`{ int x = 10; }`
A) Compile Error (Duplicate Variable)
B) Runtime Error
C) Valid code
D) x becomes 15

58. Ternary Operator type promotion:
`Object o = true ? new Integer(1) : new Double(2.0);`
`System.out.println(o);`
A) 1
B) 1.0
C) 2.0
D) Type Mismatch Error

59. Array declaration legality: Which is VALID? A) `int[5] a;` B) `int a[] = new int[5];` C) `int[] a = new int[];` D) `int a() = new int[5];`
60. What is the result of `System.out.println(10 + 20 + "30" + 40 + 50);`? A) 1020304050 B) 30304050 C) 303090 D) Error

"Part 4 and 5. The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response."

Question

61. What happens when you try to store an incompatible object type in an array?
1. The compiler throws a syntax error
2. ArrayStoreException is thrown at runtime
3. The object is automatically converted to the array type

What is the correct answer?
62. Regarding method overriding with exceptions in Java:
1. The overriding method can throw broader checked exceptions than the parent
2. The overriding method can throw narrower checked exceptions than the parent
3. The overriding method can throw any number of unchecked exceptions regardless of parent

Which statements are true?
63. Consider the following code:

```
class A { int i = 10; }
class B extends A { int i = 20; }
A obj = new B();
System.out.println(obj.i);
```

1. This will print 20
2. This will print 10
3. Variable hiding occurs, not overriding

What is correct?
64. In Java, when catching multiple exceptions:
1. FileNotFoundException can be caught after IOException
2. IOException can be caught after FileNotFoundException
3. Both can be caught using multi-catch with
65. What is the default value of elements in an int array that is declared but not initialized?
1. null
2. 0
3. undefined

Select the correct answer.
66. Regarding constructors and inheritance:
1. Constructors are inherited to subclasses
2. Constructors are not inherited but are executed when instantiating a subclass
3. Static initialization blocks are inherited

Which is correct?
67. What is the output?

```
try {
int[] list = new int[10];
System.out.println(list[10]);
} catch (ArithmeticException ex) {
System.out.println("Arithmetic");
} catch (ArrayIndexOutOfBoundsException ai) {
System.out.println("ArrayIndex");
}
```

1. "Arithmetic" is printed
2. "ArrayIndex" is printed
3. Compilation error

What happens?
68. Which statement about Java arrays is FALSE?
1. Array length is a property, not a method
2. Arrays in Java are zero-indexed
3. Array size can be changed after initialization

Identify the false statement.
69. In polymorphism, when is method overloading determined?
1. At runtime
2. At compile time
3. At execution time

Choose the correct option.
70. What happens when you catch an exception that is not declared in the throws clause?
1. Compilation error occurs
2. Code compiles successfully for unchecked exceptions
3. Runtime error occurs

What is true?

71. In a multi-level inheritance, the last subclass inherits:

1. Only methods from its immediate parent
2. Methods and properties of all classes above it in the chain
3. Only static members from all parent classes

What does it inherit?

72. Which is NOT a valid way to declare a two-dimensional array in Java?

1. int[][] arr = new int[3][3];
2. int arr[][] = new int[3][3];
3. int[][] arr = new int[][];

Identify the invalid declaration.

73. Regarding final, finally, and finalize:

1. final makes variables constant
2. finally block executes regardless of exceptions
3. finalize is for cleanup before garbage collection

Which are correct?

74. What does ArrayList.size() return compared to array.length?

1. size() returns capacity, length returns element count
2. Both return the same value always
3. size() returns current elements, length returns total capacity

Which is true?

75. Consider:

```
int[] arr = {1, 2, 3};
int[] arr2 = new int[]{1, 2, 3};
```

1. First syntax is valid
2. Second syntax is valid
3. Both are valid initialization methods

What is correct?

76. Which of the following is a RuntimeException?

1. IOException
2. IllegalAccessException
3. ArrayStoreException

Select the runtime exception.
77. In method overriding, which access modifier rule applies?
1. Overriding method can have more restrictive access
2. Overriding method must have same or less restrictive access
3. Access modifiers don't matter in overriding

What is the rule?
78. What is the time complexity of accessing an element in an array by index?
1. O(n)
2. O(log n)
3. O(1)

Select the correct complexity.
79. Can a subclass access private members of its superclass?
1. Yes, if in the same package
2. Yes, regardless of package
3. No, private members are not visible to subclasses

What is correct?
80. What happens if an exception is not handled in a program?
1. The program continues execution
2. It propagates up the call stack and may terminate abnormally
3. Java automatically handles it

What occurs?

Question

81. What is the output of the following code?

```
Integer a = 127;
Integer b = 127;
System.out.println(a == b);
Integer c = 128;
Integer d = 128;
System.out.println(c == d);
```

1. true, false
2. true, true

3. false, false

What is printed?
82. Consider this tricky static initialization code:

```
class Test {
static { initialize(); }
private static int sum;
private static boolean initialized = false;
private static void initialize() {
if (!initialized) {
for (int i = 0; i < 100; i++) sum += i;
initialized = true;
}
}
public static int getSum() { initialize(); return sum; }
}
```

1. getSum() returns 4950
2. getSum() returns 0
3. The static block executes before initialized is set to false

What is correct?
83. What happens with method overloading when passing null?

```
void foo(Object o) { System.out.println("Object"); }
void foo(String s) { System.out.println("String"); }
foo(null);
```

1. Prints "Object"
2. Prints "String"
3. Compilation error - ambiguous call

What occurs?
84. What is the value of Double.MIN_VALUE compared to 0.0?
1. Double.MIN_VALUE is negative like Integer.MIN_VALUE
2. Double.MIN_VALUE is the smallest positive value
3. Math.min(Double.MIN_VALUE, 0.0) returns Double.MIN_VALUE

Which statements are true?

85. In inheritance with static blocks, what is the execution order?

```
class Parent { static { System.out.println("P"); } }
class Child extends Parent { static { System.out.println("C"); } }
new Child();
```

1. Prints "PC"
2. Prints "CP"
3. Only "C" is printed since static blocks aren't inherited

What is the output?
86. What happens with integer overflow in this calculation?

```
long l1 = 1000606024365L;
long l2 = 1000606024365;
```

1. Both produce the same correct result
2. l1 is correct, l2 overflows during int calculation
3. Both overflow

Which is correct?
87. When does the finally block NOT execute?
1. When a return statement is in the try block
2. When System.exit() is called
3. When an exception is thrown in the try block

When does finally NOT run?
88. What is the initialization order with inheritance?
1. Parent static → Child static → Parent instance → Parent constructor → Child instance → Child constructor
2. Child static → Parent static → Child constructor → Parent constructor
3. Parent constructor → Child constructor → Static blocks

What is the correct order?
89. What is the result of this ternary operation?

```
Integer i = 10;
int result = (i == null) ? 0 : i;
```

1. result = 10 as an int

2. result = 10 as an Integer (boxed)
3. NullPointerException at runtime

What happens?
90. Regarding private methods and overriding:
1. Private methods can be overridden in subclasses
2. Private methods cannot be overridden - it's method hiding
3. Calling obj.display() works when obj is Parent ref to Child instance

What is true about private methods?
91. What does this String comparison return?

```
String s1 = "Test";
String s2 = new String("Test");
System.out.println(s1 == s2);
System.out.println(s1.equals(s2));
```

1. true, true
2. false, true
3. true, false

What is printed?
92. When are instance initialization blocks executed in inheritance?
1. Before the parent constructor completes
2. After parent constructor, before child constructor body
3. Before any constructor in the hierarchy

When do child instance blocks run?
93. What is the result of comparing Integer objects outside cache range?

```
Integer a = 500;
Integer b = 500;
if (a == b) System.out.println("equal");
if (a.equals(b)) System.out.println("equals");
```

1. Prints "equal" and "equals"
2. Prints only "equals"
3. Prints only "equal"

What is printed?

94. In method overriding with exceptions, which is INVALID?
1. Overriding method throws narrower checked exception
2. Overriding method throws broader checked exception than parent
3. Overriding method throws any unchecked exception

Which is invalid?
95. What happens with array covariance?

```
Object[] arr = new String[5];
arr[0] = new Integer(10);
```

1. Compiles and runs successfully
2. Compilation error
3. Compiles but throws ArrayStoreException at runtime

What occurs?
96. What is Integer.MAX_VALUE + 1 equal to?
1. A larger long value
2. Integer.MIN_VALUE (overflow wraps around)
3. ArithmeticException is thrown

What is the result?
97. What does this multiple inheritance scenario produce?

```
interface A { default void show() { System.out.println("A"); } }
interface B { default void show() { System.out.println("B"); } }
class C implements A, B { }
```

1. Prints "A" when show() is called
2. Prints "B" when show() is called
3. Compilation error - must override show()

What happens?
98. Consider unboxing with null:

```
Integer i = null;
int j = i;
```

1. j becomes 0
2. NullPointerException is thrown

3. Compilation error

What occurs?
99. What is the output with instance and static block order?

```
class Test {
{ System.out.println("Instance"); }
static { System.out.println("Static"); }
Test() { System.out.println("Constructor"); }
}
new Test();
```

1. Static, Instance, Constructor
2. Instance, Static, Constructor
3. Constructor, Instance, Static

What prints?
100. What happens with overloaded methods and autoboxing?

```
void process(int i) { System.out.println("int"); }
void process(Integer i) { System.out.println("Integer"); }
int x = 5;
process(x);
```

1. Prints "Integer" due to autoboxing
2. Prints "int" - exact match preferred
3. Compilation error - ambiguous

What is printed?